

Problem 4: Divehi Numerals

Solution:

- The Maldivian duodecimal numeral system is a base-12 base-24 system, with individual words for the numerals 1 through 12 and for multiples of 24.
- The numeral digit roots are:
 - 4 → *hathar*
 - 5 → *fah*
 - 7 → *hath*
 - 10 → *dhiha*
 - The number 12 is represented as:
 - 12 → *dholhas*
 - Multiples of 24 in Divehi:
 - 24 → *fassihi*
 - 48 → *fanas*
- In general, a numeral is created by adding the suffix *-eh* to the numeral root.
- Odd multiples of 12 are obtained by combining the numeral root with “*dholhas*”.
 $[numeral-root]-dholhas \rightarrow 12 \times [numeral]$
- In general, to represent numbers larger than 12, two numerals are written consecutively so that they are summed.
 $[multiple\ of\ 12]\ [numeral\ digit] \rightarrow [numeral\ multiple\ of\ 12] + [numeral\ digit]$
- Assignment 2:
 The number 76 must be represented as (72 + 4) and not (5x12 + 4), since 72 is an even multiple of 12.

Assignment 1: Correspondence

Divehi	Number	Divehi	Number
<i>faheh</i>	5	<i>thin dholhas faheh</i>	41
<i>dholhas dhihaeh</i>	22	<i>thineh</i>	3
<i>fassihi dhihaeh</i>	34	<i>dholhas hatheh</i>	19
<i>fanas</i>	48	<i>fassihi</i>	24
<i>dholhas faheh</i>	17	<i>dholhas hathareh</i>	16
<i>hatheh</i>	7	<i>hatheh dholhas hathareh</i>	88

Assignment 2: Translate to Divehi

Number	Divehi
15	<i>dholhas thineh</i>
53	<i>fanas faheh</i>
67	<i>fah dholhas hatheh</i>
76	<i>NP</i>